ActInf Textbook Group ~ Cohort 4 ~ Meeting 14 (Chapter 6 part 2)

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hey everyone welcome back it is September 26 2023 and we're in cohort 4 meeting 14 on our second discussion of chapter six so before we jump over to the text does anyone want to just give any thought or take on six yeah briefly just Charles it's it alternates time so the earlier time now we're in cohort 4 chapter 6 and vice versa but it's all different angles on the same thing so it's not like there's something that won't make sense in six because that's cool I'll just hang around and ab carry on yeah so anyone want to give a thought on six or we can look at the questions that haven't been addressed uh yet okay let's just go to a question and then if somebody has another question they can write in the chat or raise your hand okay the authors talk about language but in language processing one usually assumes recursive processes egia context free grammar how can Fe active inference account for recursivity in my understanding hidden mark models Implement types of regular grammars I assume that hierarchical extensions I.E section 642 does not fundamentally Chang this does it did anyone here ask the question or does anyone want to give a thought on this I'm not exactly sure what the question is getting at however some senses of recursion can be accommodated by hierarchical modeling so active inference models of reading have for example letters nested within words words nested within sentences and so on so this hierarchical model embodies the um the sparsity associated with this kind of a recursive process and then another kind of related notion of recursion is closer to re-entry where something is more like self-reflexive and so that is less referring to like the hierarchically nested like kind of nested for Loops notion and more having to do with like re-entry of input and reference which we might point to first off just the fundamental re-entry of the consequences of action through perception via the niche for the agent and then a little bit more um on the sophisticated cognitive entity the recursive identity concept and like identity re-entering in reference to itself Ali go for it yeah well perhaps uh the question also um I'm not sure but U maybe one angle to answer this question is to answer whether uh generative models are are exclusively recursive models or not so uh I believe uh in active inference brain work and especially uh it's current formulation there isn't any uh restrictions for generative models to be exclusively recursive so they can be applied for both uh uh I mean nested and recursive models and also in some situations we can have um purely uh I mean nonrecursive or sequential models as well uh so uh yeah it it's in its current standing it is much more broader than uh only recursive kind of structure as uh hhms were dealing with uh a couple of decades ago great point and it kind of returns to the theme of chapter six and the questions that are asked in chapter six it's like is or what what kind of recursive phenomena is being desired to to be modeled and then there's sort of like what you intentionally include in your model like you know that you want to have this phenomena and you're going to engineer and and and um Implement motifs or mechanics to get this phenomena like it's your target phenomena you want to have um a phase transition associated with people like leaving a crowded room and then like you want to like kind of build that into the model super explicitly and then a little bit more indirectly would be you want to have some phenomena arising but you don't want that phenomena to be like explicitly a parameter in the model um but that's the chapter six question what do you actually want to model what system of interest do you want to model and and that's non-trivial to identify the boundaries of the system and to subset what you actually want to model about it and what's interesting what questions you want to to because otherwise you're just making a map that's the territory and there's no point in doing so okay how's the active inference modeling recipe or approach to modeling similar and different from other modeling approaches that's a very good question just from what people have seen or what they feel what's what's one or more similarities or differences whether people have have done [Music] some quantitative modeling like statistics or systems engineering or whether they've done some other informal analysis approach how is or could be active inference chapter 6 recipe different yeah ali uh well I think this question um is I don't know perhaps too General uh to be answered um sufficiently or helpfully but um when it comes to active inference modeling my own understanding is that uh the main thing that sets it apart from uh perhaps all the other or many other approaches uh is um its emphasis on constructing uh the useful and appropriate generative model so uh in all the other approaches uh the representation of the world or the phenomena uh we're trying to model is not something uh that it's not on uh very clear uh probabilistic representation terms uh so to speak but in active inference uh there is this clear distinction between what ex what kind of phenomena or uh What uh maybe we can say what kind of structure uh the phenomena or um uh the the system we're trying to model encapsulates in terms of Its probabilistic Behavior Uh so uh it is much more clearer than other modeling approaches uh when it comes to the representation uh of that a specific system world or uh the phenomena of interest uh uh in relation to its generative model construction so yeah I think the main distinction uh lies exactly in uh the approach to constructing that representation thanks any anyone else feel free to raise your hand or go for it I I'll also point to recent work by Ma at all from year where they uh well designing explainable artificial intelligence with active inference a framework for transparent introspection and decisionmaking it points towards some of the motifs that make it different and which give it a semantic interpretability because when features of the system including its own introspection its own confidence on its answers or its sense making are framed in terms of probabilistic parameters that that are just what it says on the label then there is a clear sense of what structure or phenomena arising from structure of the system that we're modeling that doesn't mean that the model in hand is going to be the system we're modeling obviously that's the whole map territory fallacy map territory fallacy fallacy situation but when we have um interpretable parameters that play explicit roles in semantic basian networks then um at least we're on that path yeah I'll I'll um I'll bring that paper into I don't I'm sure it's somewhere else in in here but um Charles this is the paper of uh ramstead at all on the fallacy and just in brief commonly people talk about map territory fallacy which would be the fallacious equip equence of a map in a territory people are pretty familiar with map territory fallacy map is not the territory all that you know maps are useful some are all of that and in this paper they point to a kind of extreme case where people will look at active inference free energy principle modeling and say oh well it's just making a map so it's it's just making a map of the territory so it's not the territory so it's kind of like what's what's the issue like what what's the value at um and so uh there's a few aspects to this and uh it relates to on one side um reification of the map and the territory like you go with the map to the territory and then like you you you know that it's that the map's not the territory yet still they are used to kind of co- justify each other and then another flavor is oh well then free energy principle does nothing if it's just helping us build Maps that's nothing that's just a mapping framework and that's like a fallacy on the map territory fallacy OE or or anyone else what what what other angles are there I I can actually just chime in quickly because it relates to the one or sort of two-part question comment that I did put inota last week um relating to chapter one but it had it maybe it's a Sim another angle on the same thing which is the modeling versus action it's kind of a can of worms but I don't know if you if that's enough of a prompt to kind of address these terms yeah how do you see it well I I'm not sure this is just was my one of my initial questions that came up right away um where I guess the answer is in the definitions but but in practice how is the modeling modeling as action compared to modeling in contrast to action I guess that's the best I can put it right now that's very interesting like modeling is an action it's an iterated process that a modeler takes in fact we could even do a chapter six on chapter six and make a cognitive model of somebody applying active inference to modeling eventually I expect that that would be relevant so there is a sense in which modeling is an action it's a practice it has explicit and implicit skill associated with it so modeling is action however modeling throwing a baseball isn't throwing a baseball there there's a tiny bit more I can add which I I see I had written also I think in Koda um under Cod word five of of course observing and documenting are both also action however the systems and be behaviors being observed Obed would be the primary actors in the sense of the action uh yeah anyway it has to do with that observing as action is relates to that too thank you yeah we'll we'll um return to these but but um think about the setting of of so let's just say we're chapter sixing a baseball game so we have a person who's a spectator they are still taking action with their iade to look at different parts of the field reducing their uncertainty and then covertly internally they are deploying attention as action attending to different parts of their experience so even a seemingly passive perhaps overtly passive Observer is active at least in the attentional covert internal sense if not also in the epistemic foraging icade sense like choosing where to look but none of that activity happening in The Spectator is the action happening on the field and then to connect this back to the map territory fallacy fallacy no matter how deep we go into this um model of like a spectator and an athlete it does the end of the process isn't going to be like a material spectator and an athlete the map of that setting that cognitive setting isn't the territory that's so so now now now we're immune to the map territory fallacy map is not the territory but then what is it it's a map and then um I I'll also note that that ramstead and Sak devel and friston go a little bit further than um just naming their critique of this deployment they come back even stronger and say actually free energy principle is the ideal model and any model that doesn't conform to the free energy principle with respect to the map territory scenario is either saying too much or too little Oli uh so there's this heated debate in uh the philosophy of mind and um I don't know the philosophy of Neuroscience about a mological fallacy um which uh says something to the effect that uh it is a fallacy to ascribe uh a psychological action to a part of a cognitive agent uh so when we say that uh the brain does something it's not actually what brain uh that does that specific action but uh the whole agent that uh possess U uh the brain is responsible for the action so uh I'm not sure uh how I mean in this kind of self referential uh scenario of modeling uh this mological fallacy uh holds or not or whether at what point uh we can uh escape this mological fallacy but uh when it comes to uh map territory fallacy fallacy my understanding is that the F and active inference escape this exact mological fallacy right from the beginning by not uh uh conflating map uh and the territory and uh or in other words by taking a totally different approach to the concept of agency uh uh and it's not something that just metaphorically uh ascribed to part of the system so uh when they say that uh a part of a system looks as if it does some kind of inference or um any action uh it's not just a pure metaphor it's uh built into uh the very fabric of active inference Theory so uh yeah yeah I believe in that case uh this kind of self referential scenario can be applied indefinitely if you wanted to with uh this specific kind of approach thank you Ali I'll see if this is kind of another piece of it when we say a system looks as if it is bounding surprise or it's reducing surprise that's not metaphorical or the realist Claim about territory it's a narrow and limited Claim about the map which and and now all claims about Maps unless they're being considered in some abstract mapping setting are on the foundation of the relationship between the map and the territory how is it being approached differently in actim we have figure 9.2 9.1 the metab basian so here in the center we have the cognitive model of the system of interest of the subject of mod the modeling subject and then outside of the subject object is the the this is figure and ground so we're being explicit that we're making a map about the cognitive about the cognitive model of the subject now that subject's cognitive model could be a map of the environment as well but once we enter into this mode of modeling we're not making claims about the subject we're either making claims about our map of the subject and then a subset of that is maps that the subject makes makes about environment I hope that didn't say too much or too little but I think that is what what what it resonates in me when when Al say is what we kind of head off that fallacy from the beginning because the task that we're doing in cognitive modeling if this is a rat in a teamas or if this is some you know digital environment or something like that is we set up our work explicitly as making a map rather than like well what does the amydala do it's like well then you're going to get into a situation like are we making claims about the amydala because that's a claim about the territory but if we say well we're building a map of the Amiga and one of the functionalities in our map is that the Amiga is making a map of its environment so that's a kind of nested mapping and that's this that may related to the earlier recursion question but now if we have um pre prevented or styed off this map territory fallacy which is actually fairly easy to identify if it's approached hygienically then we're just not making claims about the amygdala directly we're building the map which can include submaps but then there doesn't have to be I I mean this is the uh the the Sila in the cibus map territory fallacy on one hand and then the fallacy fallacy on the other hands because like the map's utility in relevance is related to it's its mapping of course which we we want to be able to say things like yeah Fourth Street and Fifth Street Run in parallel and I'm not just talking about my map but that that's the empirical um I had a question about the map and territory could you explain uh more about what the fallacy is yeah I I'll give it first take broadly the map territory fallacy is when a constructed map which is to say an abstraction or a model of some real material system of Interest or territory when especially in a science or research or modeling setting it can be easy to kind of conflate or use rhetoric from one regarding the other so we have a bunch of healthcare data on you know doing this and this health outcome and we did a linear regression and we found that smoking was increasing the risk of lung cancer sounds like a relevant thing to say but the more narrow statistical claim would be there was a statistically significant positive regression slope between this measure and that measure so that's on the mapping we did a linear aggression and we want to be able I mean it's like look at the line don't we want to be able to say that there's like a positive relationship between these things in the real world so that's the tension because you'd want to be able to just first off look at your statistical results on the map and make a statement about the territory like doing this increases um but that's already leaving the map it's like you're looking at the map it's like well it looks like these two roads intersect um at this GPS coordinate but it's like but that doesn't mean that they do especially if you're like extrapolating so so then if somebody were to have smoked twice as much then they would have this much more so the map territory fallacy is just it's done in science communication it's done in selft talk it's done in so many different settings where something is gleaned or constructed in the map as all maps are and then that is used to justify and sometimes even linguistically is is woven or completely mixed into claims territory which may be valid or aligned or resonant or like pointing in the same direction so it's not like we're not in a situation where like maps are like deceptive unless they are but that is the tension what can you say about the map and the territory but it starts with just the fundamental recognition and and the repeated exhortation that the map's not the territory which is why you hear that as well as like all models are wrong some are useful or the borest story of the the map maker who like makes a map that's so detailed that it covers everything and then like it becomes the territory those are all kind of allegorical ways to talk about it but the big issue is when we're doing scientific modeling we're doing map making and there's this complexity of relationship between any map and any territory and again it would be so simple and straightforward to just make the map do the statistics and then the statistics should just apply right back to the territory you know I have GPS locations on these two objects I made a statistical model of their location in the statistical model they are close so can we say that the objects are close it's like just let me say that but if it were that way yeah exactly as Charles says it's not so clean distinct there are settings where it is clean and distinct but those are not every setting that makes sense and it happens a ton in Neuroscience as well because the territory is so messy and ineffable that it's like somebody will make some list of terms or they'll make some axes that they're studying summary statistics or um personality traits or just somebody brings some framework to to reckon or to grasp with Neuroscience or with psychology and then they start reifying the existence it's like I can't believe there's things in the upper left quadrant of this thing that I made but that's again it's a statement of maps on territories um aith or oi sorry go ahead uh yeah I just wanted to mention uh a probably useful distinction made by uh delus between representation and uh cartography so uh he makes a clear distinction between what we want to represent in other words uh what we want to trace uh in our models and uh the cartography or uh the action of mapping uh something in the model or diagramming or metam modeling or whatever uh so uh in this sense uh I think what active inference does uh I mean in light of this distinction between representation and cartography um more maps into uh the representation uh side of modeling uh in other words uh the very active constructing a generative model uh of the of the world or the phenomenon phenomenon of interest is done by tracing uh the probabilistic states uh of the world so it's not cartography in the sense of mapping or diagramming or met modeling but rather it actually does uh this very active tracing or representation so uh I think it also helps to uh distinguish between uh the usual way of uh talking about mapping or modeling in other scientific Endeavors and uh the active inference way of doing things would there ever be a case where you are talking about the territory like I'm just kind of wondering like when you would ever talk about the territory and science because wouldn't it always be sort of some representation you're working with what do you think olly yeah so uh it depends on what um what we mean by um representation in relation to the territory we're trying to quote unquote represent uh but u i mean even the simplest Act of observing the world and um I mean observing anything that we want to model uh entails uh at least uh an elementary uh kind of um representation or cartography combin combined so uh maybe uh I mean it's not possible theoretically and philosophically even to uh have a direct access uh to the territory we want to study but the only way we can um we can have anything useful to say about uh that specific territory is through the representation in cartography um at least this is uh one of the um views hold held by um structural realists uh and specifically ontological structural realists so it's not uh it's not about the content of the world but the structure uh a structure of the the content of that world uh manifests themselves that scientists are interested in uh so yeah at least in in some uh perspectives it's not possible to have that direct knowledge of the territory I think it's an excellent question as many we just kind of surfac them and it's for people to consider but but if we were engineers and if we were building a bridge and we had the blueprint and we were talking about the blueprint like is it sufficient to just make strong claims about blueprints that we have in hand on the table and then understand that there's a Rel relationship between the blueprints that we make and the territory that we care about is that enough like is that like a firewalled approach we're we're only going to talk about the blueprint that we have on the table and blueprints are related to Bridges or are there times where we want to bypass the blueprint and just say um you know maybe look at the bridge and say these two things look like they're this way um and then yeah what do you think about this yeah like I guess there are some cases where you'd have to make sure you know that they're distinct um especially like when you're considering the things the blueprint can't really capture about the bridge yeah and in in neuro and psychology these these come up especially like when classifying brain regions or when classifying um different neurod diversities one topic um that that Ali's mentioned of the traces in the world this reminded me of Bruno Lor's work on actor Network Theory and ants and this is a very like Proto active approach it's all about resuming the task of tracing and it's it's very provocative and another um I think provocative philosophical um term in this uh space is always already think like these are just two English words come on now the perception of phenomena by the mind of an observer the features of phenomena that seem to precede any perception of it are said to be always already present that is very provocative I believe or evocative it's like it was already that way so we're showing up in the Stream of things potentially resuming the tracing of associativity in a map making setting Charles quick context definition of blueprint and Bridge there I was being completely exoteric I really was talking about a civil engineering setting with people sketching out a blueprint like an architectural grounds plan about a bridge that like cars would drive over so compared to the actual bridge then yeah and then the neuroscience setting you know we'd have we have the person's body is the territory okay cool and then yeah and then the blueprint would be some some you know chapter 5 like well here's how you know the elbow bonees connected to the knee bone but the body is a complex territory which is why so much has been written and explored on it indeed thank you yeah the body's a bridge too right sorry go ahead and the body's a bridge and the body's body as map body as void you know this as that this already always as that like that's that's the fun um okay let's go to this question from Andrew okay selft talk is one of the topics or activities where map territory concerns apply any direct work or paper references on this topic you could share H what do people think and yeah just to to chime in I mean I I just heard you say that along the way whenever we were um discussing the policy there and it just really piqued my interest so um but uh I mean I can already imagine how that could play out as far as someone considering you know the themselves the cognitive process of like they're attempting to map whatever it is that's going on with them whatever it is they're directing their attention towards versus the territory that is entire generative process going on that that probably from an active inference framework they're only going to be able to kind of approximate in their own model of that territory but in any case I just um so uh feel like I could get into an infinite regress here with just thinking of what's going on whenever I selft talk and uh was curious if there was any published work here or um anything along those lines I don't know of any published work on this I'm sure there are analyses on like self-reported self-talk but it's something that is understandably hard to measure in flow and you get into all these kind of like recursive questions what if you remembered saying it to yourself but you didn't in the moment or you did say it in the moment but you didn't remember it or tend to it um yeah I I mentioned it just because it is like the kind of Singularity of map and territory like okay let's just say that the that a cognitive activity is making Maps so um just you know putting ourselves in the driver's seat not figure 9.1 not behavioral researcher but you could take that stands to but View From the Inside we are making maps of our ter you know where OB objects are in the room and all these other things and then that map reenters through her language and sub vocalized language so it is embodied action as well a so by selft talk do you mean like an inner monologue um I think I might be missing some context yeah yeah like stream of Consciousness whether thing just the the internal flow in imagination or even explicit self- coaching uh I've seen a paper published where they use like fmri data to predict what people's inner monologues are if that might be something that could be some way of mapping out this territory um also what people are sort of imagining visually and also what they're listening like what they're perceiving in terms of Music things like that um I think like the latest sort of deep learning models are actually getting sophisticated enough where they can be able to predict what people are intuiting just using like the right type data just to say that that very much is in line with the kind of thing I'd be interested in uh I guess bonus points that I'm interested in neuroimaging data especially so um I just shared is is this um I can send some papers that actually yeah put them in here just follow the little token and then add them in here oh I I just shared one just to see if I could confirm with you is is this at all in line with what you were referring to there a uh is it on the screen oh yeah yeah that is uh P FM D um just a quick Google search not trying to throw us off track here uh this is not the one I was referring to but I can find the one I was referring to I think it was like recent it was like in 2023 oh that even better right great I really appreciate it thank you wow yeah it's this is very I mean and to kind of return to chapter six like this is why we have a chapter six this is why unified approaches to perception cognition action matter this is why having an active inference ontology is important this is why having a recipe for making generative models and having some of this meta Theory or meta position on the relationship of map and territory fallacy and fallacy fallacy and fallacy cubed um because these are not just massively interactive and nonlinear systems of Interest they're also ones that were always already experiencing so if it was just a rubbe Goldberg machine that would be one thing but it's still wouldn't even be of type qualitatively of without even going too deep into phenomenology and like experience and awareness it just wouldn't necessarily be of the type that would be a recursive re-entrant cognitive system and this is why um I think the textbook ends with these lines important rights of Passage in theoretical neurobiology are trying to write down a generative model experiencing the frustration when simulations misbehave and learning from violations of your prior belief when something unexpected happens so it's like this isn't just a hot take like there's you what can you say about some of these settings that's just like oh okay it's it's it's you don't just like drop the mic and like dust off your hands in this iterated generative modeling practice it's like especially if we're applying it to a setting like selft talk let's say or some other psychological phenomena um which you know impinges or overlaps on selft talk um you write down the generative model there's a recipe it's chapter six plus some of the other pieces that people have added in you write it down you roll it out you see what happens did you expect what happens well if you understand act in and you made a simple example you may have you may expect get exactly what you expect back like well I made it I made an oscillating pendulum and it it did exactly that it's like great well that that might be like a homework assignment for like a certain course but if we're interested in something with more richness than like a pendulum then when we unroll that generative model we will be surprised and then what do we do with surprise well we know there's two ways to bound it change mind change the world change your mind learn an update change the world re-enter change the generative model itself on the table so having a structured approach profession ontology methodology toolkit etc etc comes into play and chapter six is kind of the transition point or like the fulcrum that the book is actually organized around chapters 1 through three get us to active inference context low road high road to or of chapter 4 modelss chapter 5 generative models in the Maman Neuroscience setting and then pretty much as soon as we can chapter six now it's us making that generative and even without extensive math or programming experience we can start to address some of these questions and then we're already in the flow of things it can definitely feel like chapter 4 is like well chapters two and three can feel like waiting through like every single Department of learn in just to get two and then chapter 4 can feel like getting through is like impossible slash improbable all totally make sense in terms of their evaluation and how they're experienced but then like we break out on the other side of four just from a page Turner perspective in six and this is like such an incredible open conversation about modeling and these qu they're hard questions to talk about but hopefully people can see that at the single entity or at the research group group scale having alignment or shared reference frames around some of these topics are make or break thanks ath you can add just just you know add make little sub Pages or add your own question or just you know just just the more references and the more things people add the better it's awesome okay yeah I can do that thank you one other piece I'll just add on this kind of map territory is the the digital introduces a very interesting setting that's kind of like a chessboard in the sense that it's fully observable and that these are virtualized where strong concordances between maps territories can be constructed to exist so like there can be a schema an abstract pattern or scheme that then generates the territory in a we know ultimately it's embodied through like thermodynamics and information all this so so not but but to a first approximation the digital systems have a different map territory relationship than Birds to the inner talk Point body's eye this just made me think like inner talk is is re-entrance auditory action and perception even though it doesn't go through the ear I assume that these um Imaging studies ask the question maybe even find the result that there's like activation of of auditory processing or something like that um then we have visual but these are so those are um internalized exteroceptive processes inner eye inner ear inner touch but then like do we have internalized interceptive as well like can we take an inner virtual view on something that's inner and and is that you know different where is it more or less relevant than selft talk but if you could take an inner position on the interpretation of the selft talk then I think we get into Andrew's infinite regresses yeah I don't know it just brings up so many questions for me of how to conceptualize that one because I mean in in a certain way if someone were to engage in selft talk I could view it is like they're if we view the self itself as like a kind of M map that the the person has constructed so you're you're you you've taken your map and and now you're making a map of your map perhaps like analyzing it or maybe you're just engaging in some kind of nested structure learning like oh let me revise my model of myself I I don't know there's so many I think a you know some kind of more empirical work would need something a little bit more finessed than the way I'm talking about it right now but it's they interesting ways it could go I could imagine know yeah like the system can't a a system cannot provide if it if it just represents everything then it's just a pass through system there's no attention or distillation or representation just a straight pass through so whether we're talking about ourselves or some other hypothetical system the self-representation that it explicitly makes to itself is a sample or a selection or an attention distribute reallocation of something else like that's just like kind of light dark like I is it all equally lit and just a pass through or is there going to be some area that is brought more into focus in the representation and then so what should be represented maybe that's where people have different um views obviously but there's the preference to have our representation be amicable or like Pleasant that's clearly one drive another Drive is for meaning all these different things and I think you recall somewhere earlier in the textbook uh first will the author say something about a a an agent having a model of itself in order to differentiate itself from its environment I.E the generative process around it and how that may be an essential possibly an essential aspect at least for certain entities in maintaining something along the lines of homeostasis that is if you don't have a sense of what you need and what you need to do which is part of you then then potentially you dissipate you know into the entropic uh processes around you so anyway I don't want to keep but interesting yeah it's fun yeah as as usual another great um chapter six Journey chap chapter six is is like an inflection point in the bur and and in how we approach things so we will then in the coming weeks with the alternating times head into chapter seven and eight on discreet and continuous time models chapter nine on datab based analysis chapter 10 wrapping things up all right thanks everyone till next time you bye